Anshuman Sinha

: github.com/anshunino
: linkedin.com/in/ansh-sinha
Phone : +33-745670673
: anshuman.sinha@college-de-france.fr

Education

PhD in Physics Paris, France
- Collège de France, Université Paris Cité Nov 2024 - Nov 2027 (exp.)
  Supervisor: *Dr. Hervé Turlier*
  - **Lab**: Multiscale Physics of Morphogenesis (↗)
  - **Thesis Title**: Forward and inverse modelling of cleavage patterning in early embryos

Masters M2 in Mathematics, Vision and Learning (MVA) Paris, France
- Institut Polytechnique de Paris Oct 2023 - Oct 2024
  - **Grade**: 16/20
  - **Relevant Courses**: Deep Learning in Practice, Reinforcement Learning, Computer Vision, Robotics

Bachelor of Technology in Electrical Engineering Surathkal, India
- National Institute of Technology Karnataka (NITK) Jul 2018 - May 2022
  - **Major**: Electrical Engineering; *GPA: 9.16 /10*
  - **Minor**: Information Technology; *GPA: 9.00 /10*

Research Experience

Project Associate Bangalore, India
- Indian Institute of Science (IISc) : Visual Information Processing Lab June 2023 - Sep 2023
  Supervisor: *Prof. Rajiv Soundararajan*
  - **Generative Image Quality Assessment**: Designing deep learning methods to determine the quality of images produced by Generative AI models.

Visiting Master's Valorization Intern Lausanne, Switzerland
- École Polytechnique Fédérale de Lausanne (EPFL) : Biomedical Imaging Group Oct 2022 - Apr 2023
  Supervisor: *Prof. Michael Unser*
  - **Steerability in 3D for Orientation Estimation**: Using steerable filters with spherical harmonic bases for one-shot 3D orientation estimation in continuous domain.

Bachelor's Thesis Cambridge, UK
- European Bioinformatics Institute (EMBL-EBI) : Uhlmann Group Dec 2021 - Sep 2022
  Supervisor: *Dr. Virginie Uhlmann*
  - **Deep Reinforcement Learning for Active Contour modelling**: Theorised and designed algorithm to detect instances in bio-images by formulating continuous control on spline models as an RL paradigm.

Research Intern Remote
- Carnegie Mellon University (CMU) : Xu Lab July 2021 - Sep 2021
  - **Unsupervised Learning**: Implemented PUB-SalNet and Double U-Net to analyse 3D cryo-ET images of sub-cellular tomograms for unsupervised classification of particles on unlabelled data.

Publications

- S. Ichbiah, **A. Sinha**, F. Delbary, H. Turlier* (2025). "Inverse 3D microscopy rendering for cell shape inference with active mesh" arXiv:2303.10440. Accepted to the: *International Conference on Computer Vision (ICCV) 2025.*

Work Experience

Research Intern Paris, France

- R&D Vision April 2024 - Oct 2024
  - **Data Augmentation and Synthetic Data Generation**: Analysing the effect of different data strategies based on generative techniques to augment low volume datasets with a focus on railways and agricultural data.

Software Intern Bangalore, India

- Texas Instruments May 2021 - July 2021
  - **Data Analytics**: Generated metrics and analysis for temporal and spatial performance of 500+ Deep Learning models trained in-situ on physical chips. Additionally developed algorithm to automate memory optimisation.
  - **Web Development**: Designed a cached data-intensive and interactive visualisation tool catering to raw multimodal data parsed directly from hardware.

Skills

Technical Skills:

- Operating Systems: Windows, Linux
- Programming Languages: Python, Matlab, Java, C
- Tools: Git, PyTorch, TensorFlow, MySQL, Streamlit

Proficiency in English: IELTS Band 8.5/9.0

References

Dr. Hervé Turlier

- Principal Investigator, Multiscale Physics of Morphogenesis
  CNRS Researcher, Collège de France
  - **Email**: herve.turlier [at] college-de-france.fr

Dr. Virginie Uhlmann

- Director, BioVisionCenter at University of Zurich
  Visiting Group Leader, European Bioinformatics Institute (EMBL-EBI)
  - **Email**: virginie.uhlmann [at] uzh.ch / uhlmann [at] ebi.ac.uk